10/14 Probability & Statistics

Gen Definitions / Jargons

Population: Entire group of items we are interested in
Sample: A subset of population.
Statistic: A number that summarizes some aspect of a group of items (mean, standard deviation, ... etc)

Population Statistics for X

Mean: $\mu_X = \frac{1}{N}\sum_{i=1}^{N} X_i$

Variance: $\sigma_X^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu_X)^2$

Standard Deviation: $\sigma_X = \sqrt{\sigma_X^2}$

Sample statistics for X

Mean: $\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$

Variance: $s_x^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$

Standard deviation: $s_x = \sqrt{s_x^2}$

Median of X

$$x_m = \begin{cases} X_{\frac{N+1}{2}}, & N_{odd} \\ \frac{X_{\frac{N}{2}} + X_{\frac{N+2}{2}}}{2}, & N_{even} \end{cases}$$

example:

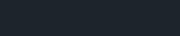
$A = \{a_1, a_2, a_3, \ldots, a_N\}$

$A_m =$

We generally want to know the statistics of a population

- If unfeasible, a sample is taken with the "hope" that it represents the population

R: Visualizing data

- Boxplots
  Useful for visualizing data distributions
  Separates data into 4 parts
  
- Histograms
- Scatter Plots
  Includes line plots